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IARCO RESEARCH PROPOSAL

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Research Topic: Assessing the impact of HIV-Subtype on the replicative fitness cost of the M184V, L74I under Lamivudine-containing Antiretroviral Therapy.

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Introduction

HIV-1 remains one of the most enduring challenges, causing detrimental effects towards humanity (especially during the 1980s) through developing into fatal AIDS. To this day, antiretroviral therapy (ART) still fails to fully eliminate the virus, allowing constant evolutionary standoff towards the drugs. This can be attributed to the lacking proofreading ability of HIV's reverse transcriptase enzyme [1], facilitating the virus to generate high rates of genetic mutations and eventually allowing the virus to develop resistance to specific drugs. For example, the ART including lamivudine drives the emergence of M184V across HIV-1 subtypes [5], while RT's error-prone replication allows the random emergence of L74I, which persists through positive selection due to its role in partially restoring the replicative fitness cost imposed by M184V [3][8].

Research Problem

In this proposal, I will focus on 2 subtypes of HIV-1: Subtype B, C and recombinant form CRF01_AE. For each type, they have a possibility of M184V, L74I emerging [3]. Thus, we can assess how the development of M184, L74I from the lamivudine-containing regimens could influence the replication rate (fitness) of the subtype. By understanding the fitness cost for each scenario, we will be able to recommend whether we should or not use treatment with lamivudine as an approach. For example, if Subtype B, containing both M184V and L74I, has a higher replication rate compared to the wildtype (prior treatment), we could recommend not using antiretroviral therapy.

The problem behind the high fitness of HIV-1 is because it could lead to a progression to AIDS [7]. Higher fitness means that they're more rounds of replication per day [5], leading to infecting more CD4+T-cells – lymphocyte with CD4 protein receptor which fits with HIV's protein gp120, allowing viral core of HIV to enter the cells and eventually kill the cell [6]. The infection of CD4+T-cells is crucial because it could weaken the immune system, allowing progression to AIDS. Therefore, research and acknowledging the fitness cost for each subtype will be important in the judgment in using lamivudine as an approach or not.

Research Questions and Hypothesis

"How does the presence of M184V alone and M184V alongside L74I differentially affect the replicative fitness of HIV-1 subtypes B, C, and CRF01_AE compared to wild-type virus" This research question is specific to HIV-1. This was to narrow down the scope of the study, allowing greater depth analysis on specific scenarios and subtypes. Since replicative fitness varies between subtypes prior to treatment [12], experiments should be conducted for each subtype separately: comparing each subtype's wild-type fitness against the fitness of viral populations carrying M184V alone and M184V alongside L74I. The independent variable would be the specific scenario of the variants and the HIV-1 Subtypes. In contrast, the dependent variable will be the replicative fitness measured by comparing with the wildtype before the treatment.

Hypothesis

- Given L74I's significantly higher rate of emergence in subtype C, confirmed to compensate for M184V's defective replication [8], I further hypothesize that subtype C carrying both M184V and L74I may demonstrate a neutral or positive replicative fitness relative to wild type.

Existing Literature

Mechanism of M184V

Firstly, I will examine how lamivudine containing ART suppresses the HIV-1 replication. Lamivudine closely resembles one of the four natural nucleotides that reverse transcriptase uses (Cytosine). Despite their different sugar rings (Lamivudine has an oxathiolane ring, Cytosine has a ribose sugar rings), they have the same structure for pyrimidine nitrogenous base [4]. Therefore, reverse transcriptase's active site acknowledges it as Cytosine, which then gets incorporated into a DNA strand. However, Lamivudine is missing a real chemical group a real nucleotide has (a 3'-OH group) which causes the DNA strand construction to pause, blocking the chain from growing any further [1].

Figure 1 – Sequence alignment confirming the ATG to GTG substitution at codon 184

```
MH575372.1      ttagatcaaagaatccagaaataactatctatcaatacatggatgacttgtatgtaggat    868
Z99328.1        TTAGAAAACAAAATCCAGACATAGTTATCTATCAATATGTGCATGACTTGTATGTAGGAT    571
K03455.1        ttagaaaacaaaatccagacatagttatctatcaatacatggatgatttgtatgtaggat    3120
                *****  * * ******** ***  ************  ******* **************
```

This is where the mutation M184V plays a role in resistance. Focusing on the 184th codon in the Reverse Transcriptase (RT) gene region within the HIV genome, mutation occurs (ATG turns into GTG). This is evident in Figure one where I compared the genetic sequence of the isolates (With and without the treatment) and the HIV gene itself in the same region (HXB2 codon 184). Patient 1's (MH575372.1) samples were taken prior treatment and patient 2's sample was taken while the patient was actively on the lamivudine-containing regimen (24th week). The change in their base pair from A to G causes the virus to contain the amino acid Valine instead of the Methionine. This is the significant change which leads to the critical adaptation to lamivudine.

Amino acid at position 184 (Methionine or Valine) occupies a specific spot in RT's active site. Therefore, depending on which amino acid it is, the shape of the surrounding pocket is affected. Subsequently, it affects how well the whole active site can accommodate the lamivudine. For example, Methionine's flexible structure does not obstruct lamivudine, allowing it to incorporate instead of the real cytosine. However, Valine's bulky, rigid structure bumps into specific parts of lamivudine and clashes with Lamivudine's oxathiolane ring, preventing Lamivudine from binding properly [4]. Thus, the mutant virus continues to replicate within the host cell, and its resistant genotype spreads through the viral population.

Fitness-cost of the M184V mutation

The same structural change caused by the mutation I discussed earlier also imposes a replicative fitness cost on HIV-1. Position 184 (where the structural change occurs) is also responsible for binding, incorporating natural nucleotides as well as excluding lamivudine. From Methionine switching to Valine, its β-branched side chain reduces the flexibility of the active side pocket and makes RT slightly less efficient at positioning natural nucleotides compared to the wild type (Methionine). This claim is supported as M184V-carrying virus has been shown to be consistently 4-8% less fit than wild-type virus in the absence of lamivudine [5]. However, since this measurement was obtained using a controlled in vitro isolate, it may not fully reflect the result within an actual patient.

L74I and its ability

The fitness cost by M184V generates selective pressure for new compensatory mutations which would not have arisen without the lamivudine treatment. An example of this is L74I. From RT's error-prone replication, random mutations develop continuously. When L74I emerges in a viral population carrying M184V, it is positively selected since it restores the replicative efficiency that M184V reduced [8] (Mechanism not characterized in existing literature). This compensatory role has been supported by clinical observations.

Gregson et al. demonstrated that L74I compensated for defective replication of M184V virus, L74I being present in 10.2 percent of patients carrying M184V but absent in those without [3].

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Difference in subtypes (B, C, CRF01_AE)

The Three HIV-1 Subtypes were selected for this investigation because M184V emerges across all three subtypes under lamivudine-containing ART [3][5]. Additionally, these three types account for majority of HIV-1 infections globally. Specifically, subtype C alone represents around 50 percent of all infection worldwide (Concentrated in southern Africa and India) [11].

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Additionally, CRF01_AE dominates Southeast Asia, and subtype B remains prevalent in North America and Western Europe [11].

Beyond geographical distribution, the probability of each subtype developing L74I with M184V varies. For example, L74I presents fewer than 8 percent in subtype B, around 12 percent in subtype C and over 14 percent in CRF01_AE viruses carrying M184V [8]. Since L74I's emergence is dependent on M184V's fitness cost [3], I suggested the unequal distribution of fitness across subtypes and their variants, motivating the investigation proposed here.

Gaps in the current literature

While M184V's replicative fitness cost is well documented, particularly in subtype B, no study has directly measured and compared this fitness cost across subtypes B, C, and CRF01_AE in a controlled experimental setting. Additionally, the quantitative compensatory effects of L74I on M184V's fitness cost have not been measured across subtypes. Since these three subtypes combined account for approximately 64% of all global HIV-1 infections [11], addressing this lack of quantitative data will inform clinical judgment on whether lamivudine-containing ART remains a viable approach against specific HIV-1 subtypes.

Methodology

This proposal draws on quantitative data from published literature to investigate existing evidence for subtype-specific fitness cost differences, including M184V prevalence rates, replication capacity measurements, and L74I frequencies across subtypes. For clear visualization of genomic evidence of the M184V mutation, a sequence comparison was conducted using publicly available HIV-1 isolates sourced from the National Library of Medicine's GenBank database with Clustal Omega and Galaxy's MAFFT tool.

Experimental Setup

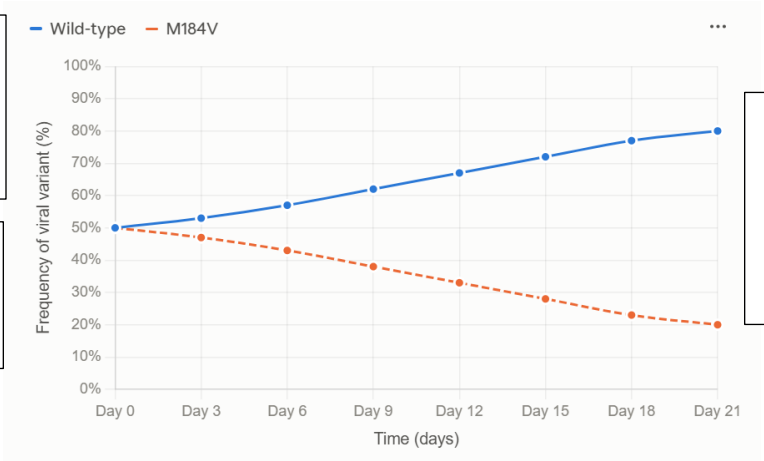
1. Prepare the isolates for three variant conditions (M184V, M184V and L74I, wild type). L74I alone is excluded as existing literature confirms it is absent in population without M184V [3]
2. MT-4 CD-4+T-cells will be cultured at 37 degrees in a CO2 incubator and confirmed viable

3. Use p24 antigen ELISA (standard HIV test that measured p24 protein concentration which provides with a precise particle count) and create a 1:1 ratio between wild type and variant (e.g. Subtype B – M184V alone vs subtype B wild type).
4. Take the mixed variant from step 3 and add it to the flask with CD4+T-cells. Collect the samples every 3 days over 3 to four weeks and store them at -80 degrees
5. Extract the RNA and reverse transcribe via Real time- PCR. Sequence using NGS (Next generation sequencing) and target the codon 184. Proportion of variant vs wildtype will be quantified at each timepoint.
6. Data will be plotted against time and fitness. Fitness would be represented as a percentage for both graphs starting from 50 percent. (Figure 2)

Figure two – hypothetical graph portraying competition between wild-type and M184V-carrying virus

Gradient of the curve represents fitness cost, steep suggests high fitness cost

Longer period for Wild Type to diverge indicates low fitness cost



There is also a possibility that the variant out competes with the wild type. Indicating enhancement in replicative capacity

Conclusion

Ultimately, by conducting this research and experiment, we will obtain specific quantitative data of fitness costs for each variant across HIV-1 Subtypes for the first time, allowing clinical decision making to become subtype aware. Currently, resistance guidelines treat all subtypes identically despite their differences. Hence, having adequate quantitative data about replicative fitness for subtypes allows informed decision on whether to maintain lamivudine containing ART in specific patients. Additionally, the result from the experiment may provide direct mechanistic explanation for unresolved patterns in existing literature. For example, L74I's disproportionate emergence in subtype C and CRF01_AE may be explained.

HIV-1 infection remains to be a global challenge and addressing this gap would not only improve clinical tools but would further encourage research about subtype specific effects on fitness. With majority of resistance testing algorithms built with subtype B data (around 10 percent of global infections) [11], the finding of this could contribute to improving the accuracy of such tools used by the remaining 90 percent of HIV-1 patients worldwide.

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